Conditional frequency distributions

May 7, 2023

[1]:
```python
import nltk
from nltk.corpus import brown
nltk.download('brown')
```

```
[nltk_data] Downloading package brown to /home/jovyan/nltk_data…
[nltk_data]   Package brown is already up-to-date!
```

[1]:
```
True
```

[2]:
```python
brown.categories()
```

[2]:
```
['adventure',
 'belles_lettres',
 'editorial',
 'fiction',
 'government',
 'hobbies',
 'humor',
 'learned',
 'lore',
 'mystery',
 'news',
 'religion',
 'reviews',
 'romance',
 'science_fiction']
```

[3]:
```python
brown.words(categories = 'news')
```

[3]:
```
['The', 'Fulton', 'County', 'Grand', 'Jury', 'said', …]
```

[4]:
```python
brown.words(categories = 'lore')
```

[4]:
```
['In', 'American', 'romance', ',', 'almost', 'nothing', …]
```

[5]:
```python
#Part 2 - Conditional frequency distributions
```

[6]:
```python
cfd = nltk.ConditionalFreqDist((genre, word)
   for genre in brown.categories()
   for word in brown.words(categories=genre))
```

[7]:
```python
#Most common token in news
cfd['news']
```

[7]:
```
FreqDist({'the': 5580, ',': 5188, '.': 4030, 'of': 2849, 'and': 2146, 'to':
2116, 'a': 1993, 'in': 1893, 'for': 943, 'The': 806, …})
```

[ ]:
```python
#Most common token in science_fiction
cfd['science_fiction']
```

[8]:
```python
from nltk.corpus import udhr
```

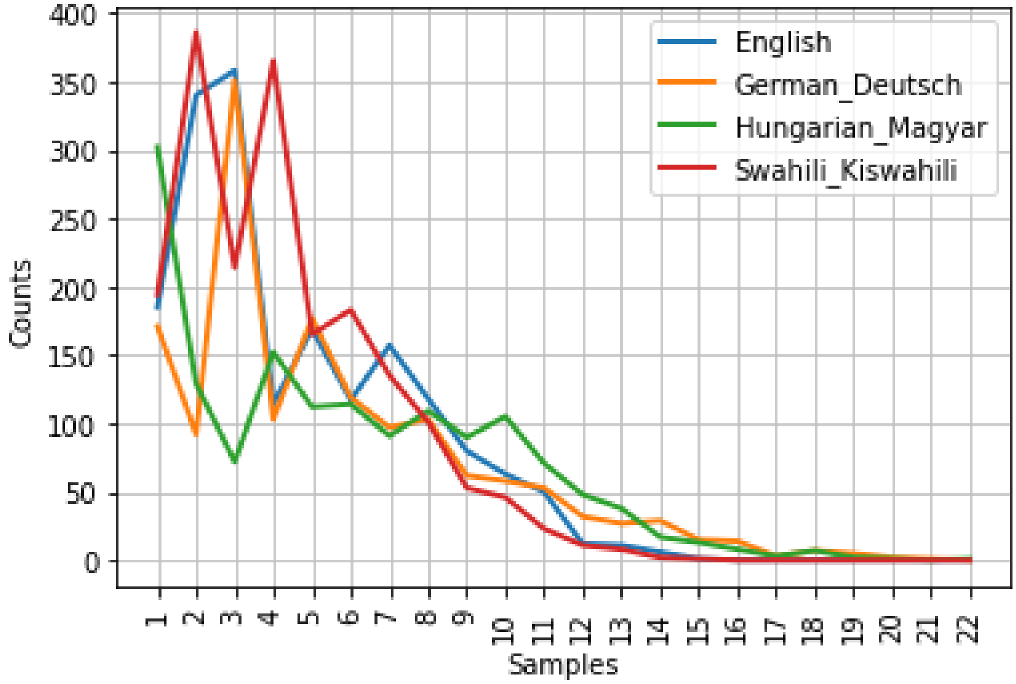
[9]:
```python
languages = ['English','German_Deutsch','Hungarian_Magyar','Swahili_Kiswahili']
```

[10]:
```python
cfd = nltk.ConditionalFreqDist(
    (lang, len(word))
    for lang in languages
    for word in udhr.words(lang + '-Latin1'))
```

[11]:
```python
cfd.tabulate()
```

```
                    1   2   3   4   5   6   7   8   9  10  11  12  13  14  15
16  17  18  19  20  21  22
          English 185 340 358 114 169 117 157 118  80  63  50  12  11   6   1
0   0   0   0   0   0   0
   German_Deutsch 171  92 351 103 177 119  97 103  62  58  53  32  27  29  15
14   3   7   5   2   1   0
 Hungarian_Magyar 302 129  72 152 112 114  91 109  90 105  71  48  38  17  13
8   3   7   2   1   0   1
Swahili_Kiswahili 194 386 214 365 165 183 135 101  53  46  23  11   8   2   1
0   0   0   0   0   0   0
```

[12]:
```python
cfd.plot()
```



[12]: <matplotlib.axes._subplots.AxesSubplot at 0x7f3899fed950>

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